CSUITE TECH LABS , COIMBATORE

A SUMMER INTERN REPORT

Submitted by:

M KARTHIKEYA

(913123106046)

In partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

Department of Electronics and Communication Engineering
Of
Velammal College of Engineering and Technology, (Autonomous)

JUNE 2025

REPORT OUTLINE

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INTERNSHIP AT

FROM : 16/06/2025 **TO : 28/06/2025**

ADDRESS : 215, BHARATHIYAR RD, NEW SIDDHAPUDUR, GANDHIPURAM, TAMIL NADU 641044

CYBER SECURITY CORPORATE
TRAINING OFFICE (CSUITE TECH LABS)

Date:15/06/2025

To

The Head of the Department,
Department of Electronics And Communication Engineering,
Velammal College of Engineering And Technology,
Madurai.

Sub: Internship Offer Letter

We are happy to permit Mr. M. Karthikeya [Reg no:913123106046], student of B.E Electronics And Communication Engineering, Velammal College of Engineering And Technology to carry out his internship in our organization from 16.06.2025 To 28.06.2025 for a period of two weeks in "CYBER SECURITY" at Coimbatore.

During this period the student will be provided the working platform with all necessary requirements by us for the successful completion of the internship. We hope this collaboration will benefit both our organization mutually.

Best regards,

For CSUITE TECH LABS

Proprietor

MR. Mani Bharathi Karunakaran
Founder & CEO, Csuite Tech Labs.
Mobile : +91 – 8838 201 976
founder@csuitetechlabs.com
info.cstlabs@gmail.com

INTERNSHIP ORDER RECEIVED FROM THE COMPANY

BONAFIDE CERTIFICATE

Certified that the student **KARTHIKEYA M (913123106046)**, completed summer internship 2025 at **CSUITE TECH LABS , COIMBATORE** under my supervision.

SIGNATURE

Dr.R.Raja Sudharsan, B.tech, M.tech, Ph.D
Assistant Professor ,
Department of Electronics and
Communication Engineering
Velammal College of Engineering
and Technology Madurai- 625 009.

Dr.P.Uma Maheswari M.E., Ph.D.,
Assistant Professor ,
Department of Electronics and
Communication Engineering
Velammal College of Engineering
and Technology Madurai- 625 009.

SIGNATURE

Dr. S. Vasuki M.E., Ph.D.,
Head of the Department
Department of Electronics and
Communication Engineering
Velammal College of Engineering
and Technology Madurai- 625 009.

ACKNOWLEDGMENT

We would like to extend our sincere gratitude to CSUITE TECH LABS for granting us the opportunity to participate in an enriching internship program focused on Cyber Security. This experience has been valuable in shaping our understanding of this complex field and equipping us with practical skills that will undoubtedly aid in our professional growth.

We would also like to express our heartfelt appreciation to Mr. MANIBHARATHI KARUNAKARAN, our industry mentor throughout the internship period. His guidance and support have played a crucial role in enhancing our knowledge and our skills in Cyber Security. His commitment to creating a conducive learning environment has made this journey both productive and enjoyable.

Furthermore, we wish to convey our gratitude to our educational institution, VELAMMAL COLLEGE OF ENGINEERING AND TECHNOLOGY, for facilitating this internship opportunity and providing continuous support throughout this endeavor.

In conclusion, this internship training has served as an exceptional stepping stone in our path toward becoming a proficient professional in Cyber Security. The knowledge acquired and the experiences gained will undoubtedly contribute significantly to our future pursuits in this field. We are deeply thankful for the unwavering support, guidance, and insights offered by all those involved.

M. KARTHIKEYA
(913123106046)

ABSTRACT

I successfully completed my internship in the Cybersecurity domain at CSUITE TECH LABS in Coimbatore, where I actively contributed to various security-related projects aimed at strengthening the digital infrastructure and safeguarding data integrity. During this internship, I was involved in the end-to-end cybersecurity process, from risk assessment to the implementation of mitigation strategies.

My responsibilities included conducting vulnerability assessments and penetration testing to identify potential security flaws across systems and networks. I analyzed threat vectors and helped in developing threat models that informed our defense strategies. I also worked on security audits and created detailed reports outlining the risks, their potential impact, and recommendations for remediation.

Using tools such as Wireshark, Nmap, Burp Suite, and Kali Linux, I gained hands-on experience in network monitoring, packet analysis, and ethical hacking techniques. I collaborated with the cybersecurity team to implement intrusion detection systems (IDS) and simulate attack scenarios to evaluate system resilience.

I participated in incident response drills, helping the team to prepare and document response strategies for different cyber threat situations. Additionally, I contributed to creating standard operating procedures (SOPs) and cybersecurity policies to ensure consistent security practices across the organization.

Regular knowledge-sharing sessions and feedback from senior analysts helped me refine my understanding of key concepts such as firewall configurations, encryption standards, access control mechanisms, and regulatory compliance (e.g., GDPR, ISO/IEC 27001). I also engaged in security awareness training initiatives, reinforcing best practices among internal teams.

Through this internship, I gained practical experience in cybersecurity tools and methodologies, enhanced my ability to identify and mitigate risks, and deepened my understanding of network and information security principles. The exposure to real-world security challenges sharpened my analytical thinking, problem-solving, and attention to detail. Overall, my internship at CSUITE TECH LABS significantly enhanced my technical skills, vigilance, and capability to contribute effectively to robust cybersecurity frameworks that align with organizational goals.

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CHAPTER-I: INTRODUCTION

1.1) DUTIES AND RESPONSIBILITIES ASSIGNED TO INTERN :

1) PROJECT DEVELOPMENT :

The intern participated in various cybersecurity-related projects, including secure networking, ethical hacking, and system audits. Key activities involved the use of real-world tools and platforms such as Threatmaps,SIM tools, and penetration testing labs to simulate and address cybersecurity threats.

2) PROGRAMMING AND SCRIPTING :

- Utilized command-line tools (e.g., ipconfig, ping, netstat, nbtstat) and scripting with Command Prompt for encoding/decoding exercises.
- Practiced scripting for Google Hacking Dork techniques.
- Applied basic Linux commands such as ls, cd, pwd, cat, ifconfig, and uname to navigate and interact with the operating system.

3) HARDWARE INTEGRATION :

- Studied and understood network devices such as routers, switches, hubs, firewalls, modems, repeaters, and load balancers.
- Explored protocols like Zigbee and Z-Wave used in hardware communication for home automation.
- Understood IEEE 802 standards and how they apply to physical and data link layer hardware integrations.

4) TROUBLESHOOTING AND DEBUGGING :

- Learned network diagnostics through tools like:
- ping, ipconfig, arp -a, and tracert for connectivity issues.
- Nmap for identifying open ports and services.
- Investigated suspicious IPs using SIEM dashboards.
- Used directory commands (dir, cd, mkdir, del, etc.) for file system management and issue tracing.

5) COLLABORATION AND TEAMWORK :

- Worked under the guidance of supervisors at CSUITE Tech Labs, contributing to group simulations in penetration testing, Red/Blue/Purple Team exercises, and threat analysis.
- Participated in Security Operations Center (SOC) simulations, including incident escalation, IP blocking, and reporting malicious activity.

6) LEARNING AND SKILL DEVELOPMENT :

Acquired hands-on experience with:

- Cybersecurity frameworks like NIST, ISO/IEC 27001, and GDPR.
- Offensive and defensive security concepts (e.g., penetration testing, incident response, digital forensics).

Improved understanding of:

- Networking fundamentals (OSI model, TCP/IP, DNS, HTTP, Subnetting).
- Cybersecurity threats and defenses (phishing, spoofing, vishing, malware prevention).
- Cybersecurity tools: Wireshark, Metasploit, Autopsy, FTK Imager.

1.2) DURATION :

- Start Date: 16/06/2025 (Day 1)
- End Date: 28/06/2025 (Day 12)
- Total Days Attended: 12 days (including both start and end dates)

1.3) BRIEF DETAILING ABOUT THE WHOLE INTERNSHIP :

During my 15-day internship at CSUITE Tech Labs, I gained practical knowledge in various aspects of cybersecurity. I was introduced to core concepts like the CIA triad, Google Hacking Database (GHDB), and vulnerability assessment and penetration testing (VAPT). I practiced ethical hacking techniques, used tools like Nmap, Wireshark, Metasploit, and worked on platforms such as TryHackMe and HackTheBox. I enhanced my command-line proficiency in both Windows and Linux environments using commands such as ipconfig, ping, netstat, ls, and ifconfig. I explored networking fundamentals including OSI model, IP addressing, ports, protocols, and routing devices. I also learned about firewalls, DMZs, and VPNs. The internship involved digital forensics using tools like FTK Imager, and incident response handling through simulated SOC tasks. I studied data privacy laws like GDPR, HIPAA, and ISO 27001, and understood threats like phishing, spoofing, and malware.Overall, this internship significantly improved my technical skills and awareness of cybersecurity best practices, preparing me for real-world cybersecurity roles.

COMPANY OVERVIEW:

CSUITE Tech Labs is a technology-driven company specializing in cybersecurity, IT consulting, and digital innovation. The organization provides advanced services such as ethical hacking, penetration testing, vulnerability assessment, and digital forensics. It focuses on helping businesses secure their digital infrastructure and comply with global data protection standards. The company promotes a hands-on learning environment, using tools like SIEM, firewalls, and intrusion detection systems. It plays a vital role in training future cybersecurity professionals through structured internship programs. Interns gain practical exposure to offensive and defensive security strategies, network security, and real-time threat analysis. CSUITE emphasizes research, continuous learning, and adaptability to evolving cyber threats. The firm supports secure network design, risk management, and endpoint protection for various industries. Its team collaborates on simulation-based exercises to strengthen cybersecurity awareness. Overall, CSUITE Tech Labs is committed to building a secure digital future through innovation and skilled talent development.

OBJECTIVES:

The primary objective of the internship at CSUITE Tech Labs was to gain practical experience in cybersecurity through hands-on training, real-world simulations, and guided project work. The internship aimed to build proficiency in ethical hacking, network security, digital forensics, and compliance practices while fostering professional skills like teamwork and problem-solving.

ACTIVITIES AND RESPONSIBILITIES:

1) TRAINING SESSIONS :

- Attended daily training on core cybersecurity topics including the CIA Triad, OSI Model, Google Hacking Database, and Vulnerability Assessment and Penetration Testing (VAPT).
- Participated in technical sessions on network protocols, ports, IP classes, firewalls, DMZ, and threat intelligence.
- Learned practical use of command-line tools (e.g., ipconfig, ping, netstat, nbtstat) and Linux basics.

2) PROJECT WORK :

- Simulated real-world attacks using platforms like TryHackMe and HackTheBox.
- Performed penetration testing on dummy targets (e.g., FakeBank simulation using Gobuster).
- Investigated security incidents using SIEM dashboards, analyzing IPs and escalating issues.

3) DEVELOPMENT TOOLS :

Used tools such as:

- Nmap – for network scanning and port discovery
- Wireshark – for packet analysis
- FTK Imager & Autopsy – for digital forensics
- Gobuster – for directory brute-forcing
- Metasploit – for exploiting known vulnerabilities

Practiced encoding/decoding using command prompt and command-line utilities.

4) COLLABORATION :

- Worked closely with mentors and supervisors in simulated Red, Blue, and Purple Team roles.
- Participated in group exercises involving incident response, malware analysis, and policy drafting.
- Engaged in discussions around cyber laws, compliance standards (GDPR, HIPAA, ISO 27001), and user awareness training.

OUTCOMES :

The internship significantly enhanced my technical knowledge and hands-on experience in cybersecurity. I developed skills in ethical hacking, digital forensics, threat analysis, and compliance management. I became proficient with tools like Nmap, Wireshark, and FTK Imager, and gained confidence in using both Linux and Windows command-line environments. The collaborative tasks improved my communication and team-working abilities. Overall, the internship prepared me for real-world challenges in the cybersecurity domain and reinforced my interest in pursuing it as a career.

CHAPTER-II ABOUT THE COMPANY

2.1) ABOUT THE COMPANY :

CSUITE Tech Labs is a forward-focused technology company that specializes in cybersecurity, digital forensics, and IT infrastructure consulting. The firm offers cutting-edge services such as vulnerability assessment, penetration testing, ethical hacking, incident response, and cybersecurity compliance support. Through a strong emphasis on hands-on training, CSUITE Tech Labs also serves as a practical learning ground for aspiring cybersecurity professionals. The organization integrates industry-standard tools and platforms to simulate real-world security environments, empowering clients and trainees to respond effectively to emerging digital threats.

2.2) MISSION OF THE COMPANY :

To provide robust, intelligent, and ethical cybersecurity solutions that protect digital assets, while fostering innovation, learning, and resilience in a rapidly evolving technological landscape.

2.3) VISION OF THE COMPANY :

To become a globally recognized leader in cybersecurity solutions and education, building a safer digital world through innovation, excellence, and empowerment of the next generation of cybersecurity professionals.

CHAPTER-III: INTERNSHIP OVERVIEW

We were assigned with the following various tasks during our internship in CSUITE TECH LABS

3.1) WEEKLY ACTIVITY REPORT :

Date From : 16/06/25 Date To : 28/06/25

DATE	TIME	DIVISION	OBSERVATIONS
16/06/2025	9.00am - 5.00pm	Cybersecurity Foundation	Covered core IT skills, hardware components, networking basics (TCP/IP, DNS, firewalls), and cybersecurity certifications. Introduced roadmap from roadmap.sh.
17/06/2025	9.00am - 5.00pm	Ethical Hacking	Explored Google hacking (dorks, cheat sheet), ethical hacking steps, AVTR model, password cracking methods, and encryption vs. hashing.
18/06/2025	9.00am - 5.00pm	Network & Web Security	Studied RSA 2048 security, OSI/TCP-IP models, unicast vs. multicast, NAT, IP types, HTTP status codes, and Google data center layers.
19/06/2025	9.00am - 5.00pm	Cyber Awareness	Focus on Royal Cybersecurity Force, 10 Cybersecurity Steps, BCP vs DCP, Web Iceberg Model, Intranet vs Internet, digital forensics tools, and BYOD policies.
20/06/2025	9.00am - 5.00pm	Virtualization And Linux	Practiced VMware setup and managing VMs, created a cybersecurity lab, and executed basic Linux terminal tasks (file, user, and network management).
21/06/2025	9.00am - 5.00pm	Windows system And CLI	Worked on CMD & PowerShell commands for diagnostics and process management, VMware configuration, and real-world system-level cybersecurity simulations.
23/06/2025	9.00am - 5.00pm	Ethical Hacking Lab	Used Kali Linux tools, set up Metasploitable/BeeBox labs, learned Metasploit basics, tested web vulnerabilities from OWASP Top 10, and practiced manual exploitation.
24/06/2025	9.00am - 5.00pm	Cyber laws and privacy	Studied IT Act (India), CFAA and ECPA (USA), understood age-specific cyber threats, and explored the ethical/legal dimension of data protection and online safety.
25/06/2025	9.00am - 5.00pm	OSI & Wireless Protocols	Explored OSI layers, IEEE 802, Wi-Fi types, Zigbee, Z-Wave, and EM spectrum.

26/06/2025	9.00am - 5.00pm	IP & Server Architecture	Detailed IP classes, DMZ zones, HTTP status codes, and handling high-traffic data flows.
27/06/2025	9.00am - 5.00pm	Networking & Topologies	Learned about routers, switches, hubs, firewalls, VPNs, and network types like LAN, WAN, and topologies.
28/06/2025	9.00am - 5.00pm	Command-Line Diagnostics	Practiced ipconfig, ping, netstat, nbtstat, and Linux commands for diagnostics and file handling.

3.2) CONTRIBUTION OF THE INTERN TO THE ORGANIZATION/ OUTCOME :

During the internship at CSUITE Tech Labs, I actively contributed to simulated cybersecurity tasks, helping identify vulnerabilities in practice systems and supporting ethical hacking exercises. I performed tasks like scanning networks, decoding encoded data, identifying weak login credentials, and analyzing IP addresses. My involvement in sessions related to phishing awareness, digital forensics, and data protection helped enhance the lab's collaborative simulations. Through consistent participation and willingness to learn, I supported the training environment and contributed to strengthening the organization's practical learning initiatives.

3.3) TASK ACCOMPLISHMENTS :

Over the course of the internship, I successfully:

- Used command-line tools (ping, ipconfig, netstat, arp -a) for basic network diagnostics.
- Practiced directory navigation and file handling using commands like cd, mkdir, dir, and del.
- Conducted Google Dorking to identify hidden web pages and data exposure scenarios.
- Participated in penetration testing simulations such as the FakeBank exercise using Gobuster.
- Explored IP tracking tools and used online platforms to determine IP locations.
- Applied basic encoding and decoding techniques (Base64, URL, and Hex) using Python and CLI.

These tasks improved my comfort with standard cybersecurity tools and laid the foundation for advanced concepts.

3.4) KNOWLEDGE GAINED FROM INTERNSHIP :

The internship provided a solid understanding of essential cybersecurity principles including network security, threat identification, data protection laws, and incident response. I learned how systems can be compromised through phishing, misconfigured networks, or weak passwords, and how to prevent such breaches using simple tools and best practices. I also understood the importance of secure system configurations, role-based access, and multi-factor authentication in modern organizations. The sessions on GDPR, HIPAA, and other data privacy regulations helped me recognize the legal and ethical responsibilities involved in cybersecurity roles.

3.5) HANDS-ON EXPERIENCE :

I received practical exposure to a variety of tools and concepts during the internship:

- Used Nmap to scan local networks and identify open ports.
- Worked with Wireshark to view network packets and understand how data travels.
- Practiced Linux commands (ls, cd, pwd, cat) for navigating files and directories.
- Used FTK Imager to view and preserve digital evidence in forensics tasks.
- Explored SIEM dashboards to analyze suspicious IP logs and raise alerts.
- Participated in phishing simulations to identify fake emails and malicious links.

These experiences provided me with a medium-level hands-on understanding of tools commonly used in cybersecurity labs and workplaces.

CHAPTER-IV: CONCLUSIONS & OUTCOMES

The internship at CSUITE Tech Labs was a highly enriching and practical learning experience that significantly enhanced my understanding of the cybersecurity field. Throughout the 15-day program, I was exposed to a variety of real-world concepts and tools used in securing digital infrastructure, conducting penetration testing, and analyzing cyber threats. I developed a strong foundational understanding of key areas such as network protocols, vulnerability assessment, digital forensics, and data protection laws like GDPR and HIPAA. The hands-on sessions enabled me to work with essential cybersecurity tools such as Nmap, Wireshark, FTK Imager, and command-line utilities, which strengthened my technical skills in a practical context. I also became familiar with basic Linux navigation, simple scripting, and Google Dorking techniques, which further broadened my skill set.

Participating in team-based simulations and task-oriented exercises helped me understand how cybersecurity operations are conducted in a professional setting. The guidance from mentors and the structured learning environment allowed me to apply theoretical knowledge to real-time scenarios, such as identifying phishing attempts, decoding sensitive data, and investigating suspicious IP addresses. Importantly, the internship emphasized the importance of security awareness, risk management, and ethical practices in safeguarding information systems. Overall, this internship not only helped me grow technically but also improved my problem-solving, critical thinking, and collaborative abilities—preparing me to pursue further education and career opportunities in cybersecurity with greater confidence and clarity.

REFERENCES

- Exploit Database – Google Hacking Database (GHDB)
https://www.exploit-db.com/google-hacking-database

- Google Hacking Cheat Sheet – AT&T Cybersecurity
https://cdn-cybersecurity.att.com/blog-content/GoogleHackingCheatSheet.pdf

- Roadmap.sh – Cybersecurity Roadmap
https://roadmap.sh/cyber-security

- J. Nielsen and R. Budiu, Mobile Usability. New Riders, 2013.

- S. Lidwell, K. Holden, and J. Butler, Universal Principles of Design, 2nd ed.
Rockport Publishers,

CERTIFICATE RECIEVED **FROM THE** COMPANY

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Nearby, Coimbatore, Tamil Nadu- 641 044
GSTIN 33CGQPM8772B1Z3

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. M. KARTHIKEYA** currently pursuing Department of Electronics And Communication Engineering from Velammal College of Engineering and Technology, Madurai has successfully completed Two Week Internship on CYBER SECURITY domain from 16.06.2025 To 28.06.2025 at **CSUITE TECH LABS, a** leading provider of innovative cyber security, AI ML Ops & Disruptive Technology solutions.

During this period, Karthikeya worked on various projects and tasks, demonstrating a keen interest in cyber security and gaining valuable hands-on experience in:

- Threat analysis and vulnerability assessment
- Penetration testing and security auditing
- Incident response and risk management
- Security best practices and compliance

Throughout the internship, Karthikeya demonstrated:

- A strong work ethic and commitment to learning and growth
- Excellent problem-solving skills and attention to detail
- Effective communication and teamwork skills

We appreciate his dedication, hard work, and enthusiasm during the internship. We are confident that the skills and knowledge gained during this period will be valuable in his future endeavors.

For CSUITE TECH L.\BS

Proprietor

Authorized Signature:
MANIBHARATHI KARUNAKARAN
FOUNDER & CEO - CSUITE TECH LABS

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